# Exponential Distribution :—

**Definition :—** A continuous r. v. X assuming non negative values is said to have an exponential distribution with parameter $\alpha > 0$, if its pdf is given by

$$f(x) = \alpha e^{-\alpha x}, \quad x \geq 0$$
$$= 0, \quad \text{Otherwise.}$$

**Note :—** $\int_0^\infty \alpha e^{-\alpha x} dx = -\frac{\alpha}{\alpha} e^{-\alpha x}\Big|_0^\infty + \int_0^\infty e^{-\alpha x} dx$

$= 1$

cumulative distribution function: $\int_0^x \alpha e^{-\alpha x} dx = 1 - e^{-\alpha x}$

$= 0.$

$E(x) = \int_0^\infty x \alpha e^{-\alpha x} dx \quad \alpha \Big|-\frac{x e^{-\alpha x}}{\alpha}\Big|_0^\infty + \frac{\alpha}{\alpha}\int_0^\infty e^{-\alpha x} dx$

$= 0 + \frac{1}{\alpha} = \frac{1}{\alpha}$

$E(x^2) \int_0^\infty x^2 \alpha e^{-\alpha x} dx = -\frac{x^2}{\alpha}\alpha e^{-\alpha x}\Big|_0^\infty + 2\int_0^\infty x e^{-\alpha x} dx$

$0 + 2\left[-\frac{x}{\alpha} e^{-\alpha x} + \frac{1}{\alpha}\int e^{-\alpha x} dx\right]_0^\infty$

$= 2/\alpha^2$

$V(X) \quad E(x^2) - \{E(x)\}^2 = \frac{2}{\alpha^2} - \frac{1}{\alpha^2} = \frac{1}{\alpha^2}.$

**Ex.** The number of years a radio functions is exponentially distributed with parameter $\alpha\ \lambda = \frac{1}{8}$. If Jones buys a used radio, what is the prob. that it will be working after an additional 10 years.

**Sol** $f(x) = \frac{1}{8} e^{-x/8}, \quad x > 0.$

$P(X > 10) = \frac{1}{8}\int_{10}^\infty e^{-x/8} = e^{-10/8} = e^{-5/4} = 0.2865$

The probability that he has to wait at least 20 min is $P(X \geq 20) = \int_{20}^{30} f(x)\,dx = \frac{1}{30}(30-20) = \frac{1}{3}$.

Ex Suppose $X$ is uniformly distributed over $[-\alpha, \alpha]$, where $\alpha$... When-ever possible, determine $\alpha$ so that the following are satisfied.

(a) $P(X>1) = 1/3$, (b) $P(X<\frac{1}{2}) = 0.7$

(c) $P(|x|<1) = P(|x|>1)$

Sol (a) $f(x) = \frac{1}{2\alpha}, \quad -\alpha < x < \alpha$

$= 0 \quad , \quad$ otherwise

$$P(X>1) = \frac{1}{2\alpha}\int_1^{\alpha} dx = \frac{1}{2\alpha}(\alpha-1) = \frac{1}{3}$$

$\therefore 3\alpha - 3 = 2\alpha$

$\alpha = 3$

(b) $P(X<\frac{1}{2}) = \frac{1}{2\alpha}\int_{-\alpha}^{1/2} dx = \frac{1}{2\alpha}\left(\frac{1}{2}+\alpha\right) = 0.7$

$\therefore 0.5 + \alpha = 1.4\alpha$

$\therefore \alpha = \frac{0.5}{0.4} = \frac{5}{4}$.

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(c) $P(|x|<1) = P(-1<x<1) = \frac{1}{2\alpha}\int_{-1}^{1} dx = \frac{1}{\alpha}$

$P(|x|>1) = P(-\alpha<x<1) + P(1<x<\alpha)$

$= \frac{1}{2\alpha}\int_{-\alpha}^{1} dx + \frac{1}{2\alpha}\int_{1}^{\alpha} dx$

$= \frac{1}{2\alpha}\left[1+\alpha+\alpha-1\right] = 1$

$\alpha = 1$.

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## Exponential distribution

The exponential distribution "lacks memory", i.e. if $X$ has an exponential distribution then for every, ~~$a>0$~~ $s>0, t>0$,

$$P(X>s+t \mid X>s) = P(X>t).$$

$$\therefore P(X>s+t \mid X>s) = \frac{P[(X>s+t)\cap(X>s)]}{P(X>s)}$$

$$= \frac{P(X>s+t)}{P(X>s)} = \frac{e^{-\alpha(s+t)}}{e^{-\alpha s}} = e^{-\alpha t}$$

$$= P(X>t).$$

$\therefore$ exponential distribution lacks memory.

Ex. (a) If $X$ has exponential distribution with mean 2. find $P(X<1 \mid X<2)$

$$[P(X<1) \mid P(X<2)] = \frac{1-e^{-\theta}}{1-e^{-2\theta}} = \cdot$$

$$\theta = \frac{1}{2}.$$

(b) If $X \sim Exp(\lambda)$ with $P(X\leq 1) = P(X>1)$. find $Var X$.

$$\therefore P(X\leq 1) + P(X>1) = 1 \Rightarrow P(X\leq 1) = \frac{1}{2}.$$

$$\therefore var(X) = \frac{1}{\lambda^2} = 1/(\log_e 2)^2.$$

(c) A Continuous random variable $X$ has the probability density function $f(x)$ given by

$$f(x) = A e^{-x/5}, \quad x > 0$$
$$= 0 \quad \text{otherwise},$$

Find the ~~prob. of~~ value of A and show that for any two positive no! s and t,

$$P(X > s+t \mid X > s) = P(X > t)$$